

Superstore Sales Data Visualization

Data Visualization using Python
DevSphere Internship Program — Week 2

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Degree	BSCS — Artificial Intelligence / Machine Learning focused student
Project	Superstore Sales Data Visualization
Dataset	Superstore Dataset — Kaggle
Primary Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

Prepared for internship submission

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Report scope. This report documents the Week 2 internship task, from dataset inspection and preprocessing through three required visualizations and the interpretation of business findings. The report uses only the project information and figures supplied for this submission.

1. Introduction

Data visualization is an important part of practical data analysis because it converts structured records into visual patterns that are easier to interpret. In this internship project, the Superstore dataset was analyzed with Python to understand sales performance across product categories and over time. The work demonstrates a complete small-scale visualization workflow: loading data, inspecting its structure, checking data quality, converting date fields, aggregating sales, creating charts, and interpreting the resulting patterns.

The project was completed as part of the DevSphere Internship Program — Week 2, under the task Data Visualization using Python. The emphasis was on producing clean, properly labeled visualizations and translating the outputs into useful business observations.

2. Project Objective

The main objective was to analyze the Superstore dataset using Python and create meaningful visualizations that communicate sales performance and business insights.

- Compare total sales across the three main product categories.
- Examine the change in total sales over the observed years.
- Show the percentage contribution of each product category to total sales.
- Practice data preparation, aggregation, visualization, and interpretation using Python.

3. Dataset Description

The Superstore dataset contains 9,994 records and 21 columns. It represents order-level business information, including customer, geographic, product, sales, quantity, discount, profit, and date-related fields.

Columns: Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country, City, State, Postal Code, Region, Product ID, Category, Sub-Category, Product Name, Sales, Quantity, Discount, Profit

Dataset source: Kaggle — Superstore Dataset.

The analysis in this report focuses on the Category, Sales, and Order Date fields for the required visualizations.

4. Technologies Used

Technology	Role in Project
Python	Core programming language for the analysis and visualization workflow.
Pandas	Data loading, inspection, datetime conversion, grouping, and aggregation.
NumPy	Numerical computing support within the Python data-analysis environment.
Matplotlib	Creation and formatting of the required bar, line, and pie charts.
Seaborn	Available visualization library for clean statistical graphics and styling.
Jupyter Notebook	Interactive environment used to document and execute the analysis.

5. Data Loading and Inspection

The dataset was loaded into a Pandas DataFrame. Initial inspection confirmed the dataset shape as (9994, 21) and displayed the available column names and first rows. This step established the structure of the data before analysis.

```
df = pd.read_csv("../data/superstore.csv", encoding="latin1")

print("Dataset loaded successfully.")
print("Dataset shape:", df.shape)
print("\nColumn names:")
print(df.columns.tolist())
```

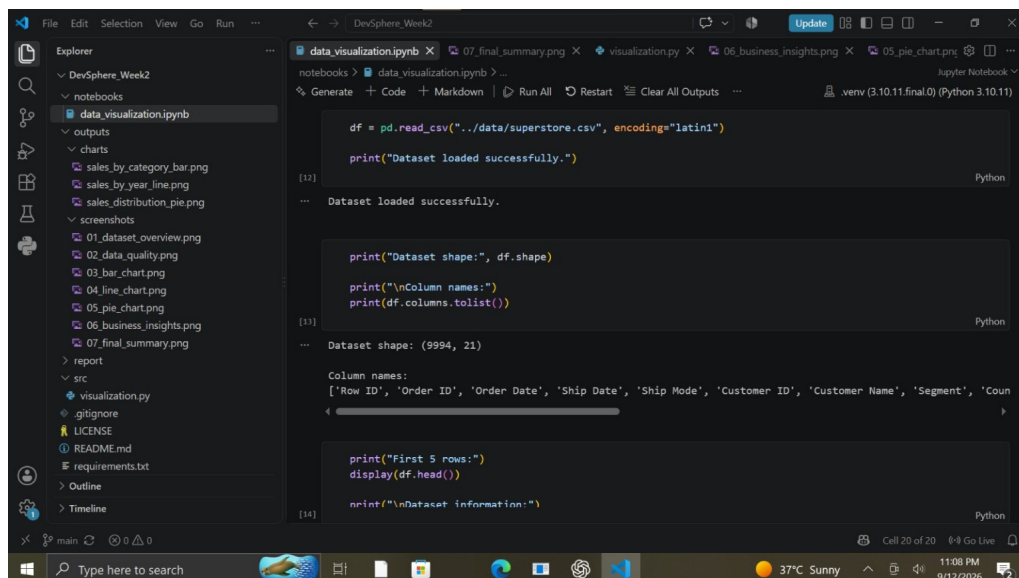


Figure 1. Project notebook screenshot showing dataset loading, dataset shape, and column inspection.

6. Data Quality and Preprocessing

The data-quality check reported 0 missing values across the listed columns and 0 duplicate rows. This provided a clean starting point for the visualization task and reduced the need for missing-value imputation or duplicate removal.

Quality Check	Result
Rows	9,994
Columns	21
Missing values	0
Duplicate rows	0

Date preprocessing

The Order Date and Ship Date columns were converted from text to datetime format using Pandas. Converting these fields was necessary for year-based grouping and time-oriented analysis.

```
df["Order Date"] = pd.to_datetime(df["Order Date"])
df["Ship Date"] = pd.to_datetime(df["Ship Date"])
```

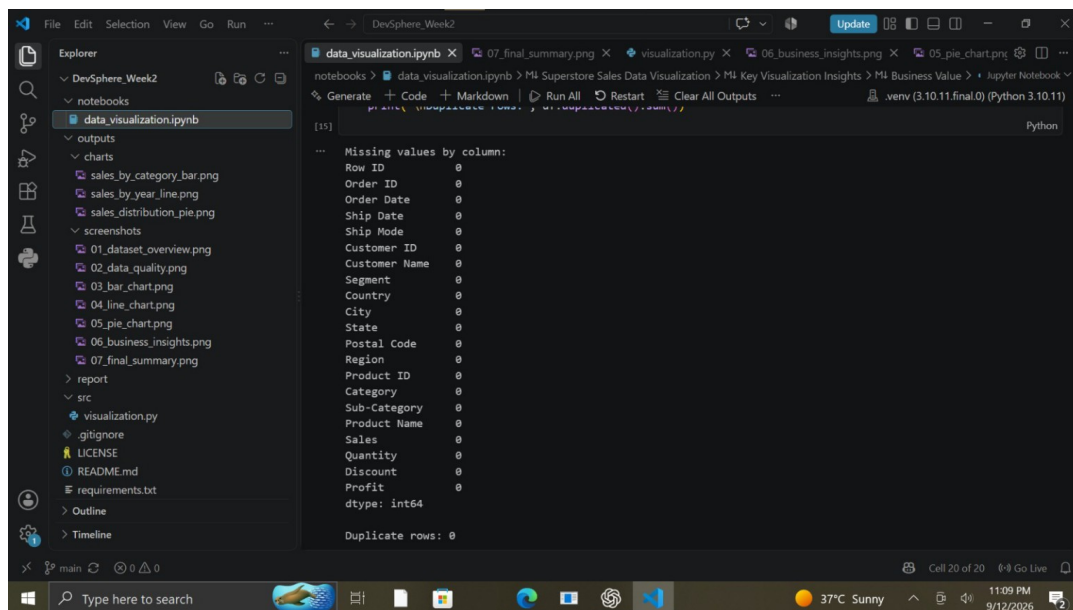


Figure 2. Data-quality screenshot showing zero missing values for the dataset columns and zero duplicate rows.

7. Bar Chart — Sales by Category

The first required visualization compares total sales across the three main product categories. A grouped sum was calculated using the Category field and the Sales field, then sorted in descending order.

```
category_sales = (  
    df.groupby("Category")["Sales"]  
        .sum()  
        .sort_values(ascending=False)  
)
```

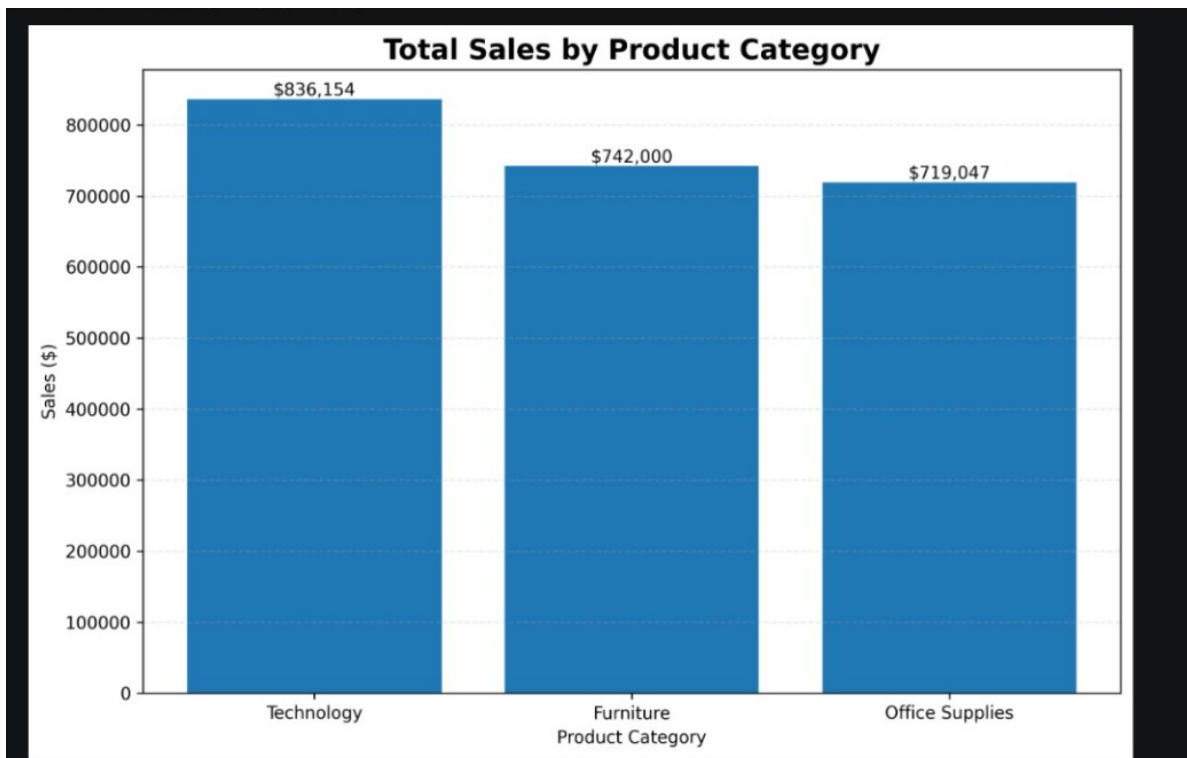


Figure 3. Total Sales by Product Category. The supplied project output shows Technology as the highest-selling category.

Interpretation

The bar chart makes the category ranking immediately visible. Technology records the highest total sales, followed by Furniture and Office Supplies. The chart also uses a clear title, labeled axes, grid lines, and data labels so that category differences can be read without relying only on the visual height of the bars.

Recorded project values: Technology — \$836,154.03; Furniture — \$742,000 (rounded label shown in the chart); Office Supplies — \$719,047.03.

8. Line Chart — Sales by Year

The second visualization examines the yearly sales trend. After converting Order Date to datetime, the year component was extracted and used to group and sum Sales.

```
yearly_sales = (  
    df.groupby(df["Order Date"].dt.year)["Sales"].sum()  
)
```

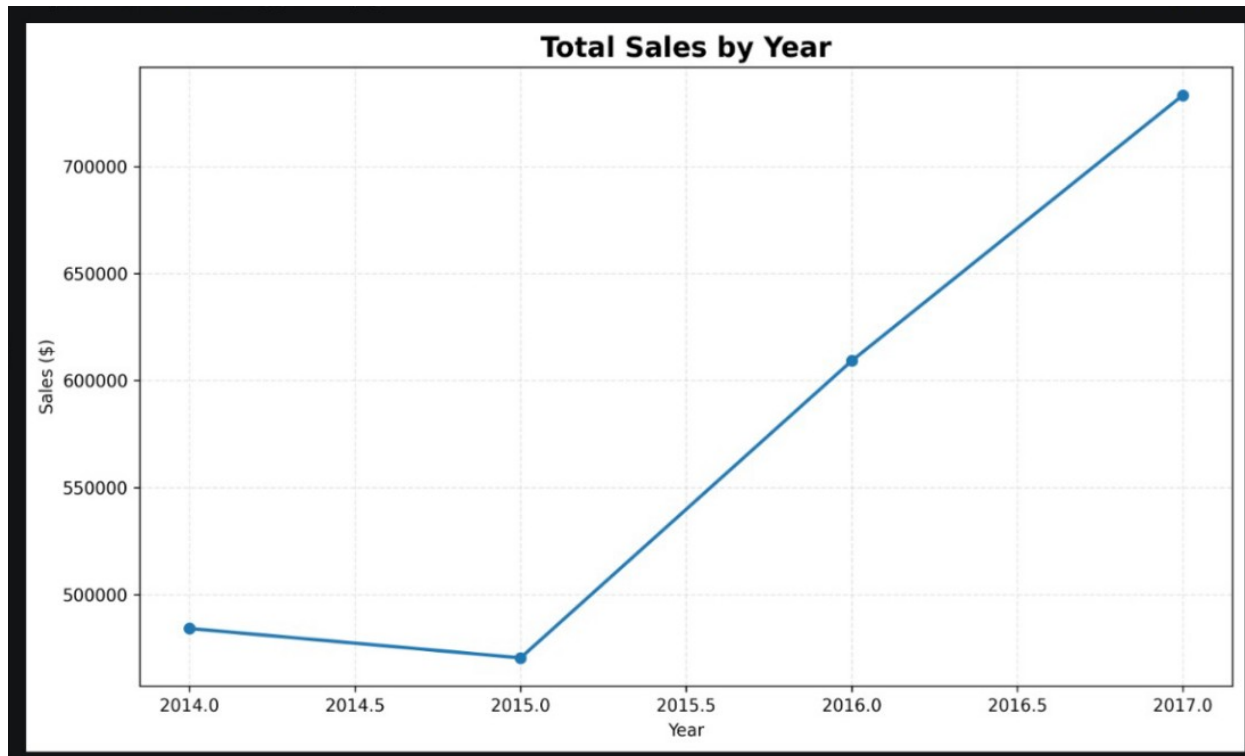


Figure 4. Total Sales by Year. The project output covers the observed years 2014–2017 and shows the overall movement in annual sales.

Interpretation

The line chart shows a small decline from 2014 to 2015, followed by a strong increase in 2016 and a further increase in 2017. The overall direction across the observed period is upward, supporting the project finding of positive sales growth over time.

The project analysis identifies 2017 as the highest-sales year, with recorded sales of \$733,215.26.

9. Pie Chart — Sales Distribution

The third visualization presents each product category's share of total sales. The same category-level aggregation used for the bar chart was used to create the distribution view.

```
category_sales = df.groupby("Category")["Sales"].sum()

plt.pie(
    category_sales,
    labels=category_sales.index,
    autopct="%1.1f%%"
)
```

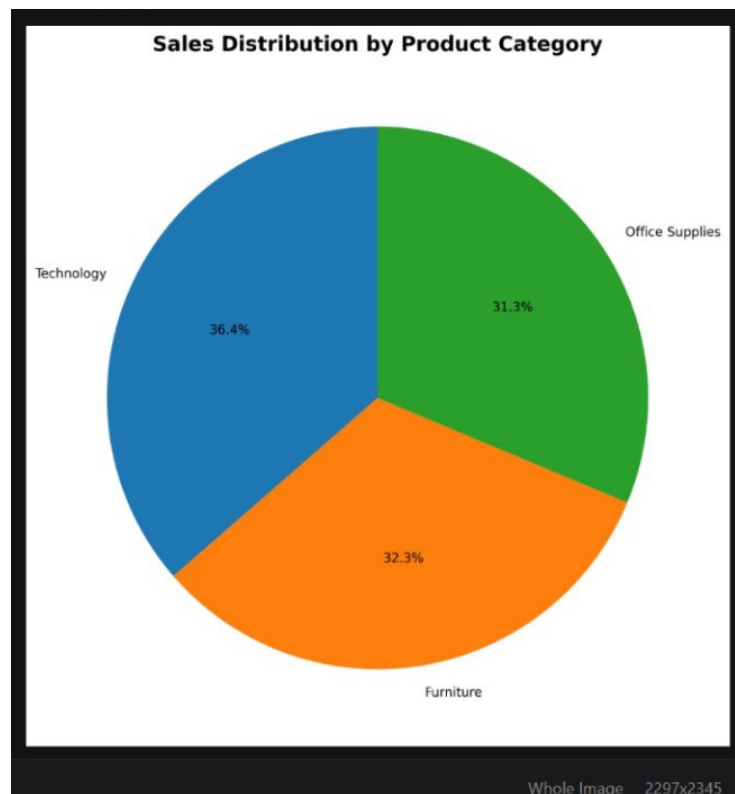


Figure 5. Sales Distribution by Product Category. The supplied project output labels Technology at 36.4%, Furniture at 32.3%, and Office Supplies at 31.3%.

Interpretation

The pie chart communicates the composition of total sales rather than the absolute category totals. Technology represents the largest share, while Furniture and Office Supplies account for substantial portions as well. The distribution therefore shows that sales are spread across all three categories, with Technology holding the largest contribution.

10. Business Insights

Area	Finding	Business Value
Sales by Category	Technology is the highest-selling category, followed by Furniture and Office Supplies.	Supports category comparison and identification of strong-performing product groups.
Sales Trend	Annual sales show an overall increase across the observed period, with 2017 identified as the highest-sales year.	Helps monitor sales performance over time and recognize changes in the overall trend.
Sales Distribution	Technology has the largest share of total sales; the three categories together form the complete sales composition.	Provides a concise view of sales composition and category contribution.

Practical business value

- Compare product categories using consistent sales measures.
- Monitor sales trends and identify periods of stronger performance.
- Recognize strong-performing categories for further analysis.
- Understand the composition of total sales.
- Support data-driven decision-making with visual evidence.

```

print(f"\nLowest-selling category: {worst_category}")
print(f"Sales: ${worst_category_sales:,.2f}")

print(f"\nHighest-sales year: {best_year}")
print(f"Sales: ${best_year_sales:,.2f}")

[20] Python

KEY BUSINESS INSIGHTS
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Highest-selling category: Technology
Sales: $836,154.83

Lowest-selling category: Office Supplies
Sales: $719,047.83

Highest-sales year: 2017
Sales: $733,215.26

import os

os.makedirs("../outputs/charts", exist_ok=True)

# 1. Bar Chart
plt.figure(figsize=(8, 5))
category_sales.sort_values(ascending=False).plot(kind="bar")

[21] Python

```

Figure 6. Project notebook screenshot showing the recorded business insights, including the highest-selling category and highest-sales year.

11. Project Structure

The project is organized into separate folders for the notebook, reusable visualization code, chart outputs, and supporting project files. The structure supplied for the project is summarized below.

```
DevSphere_Week2/  
├── data/  
│   └── superstore.csv  
├── notebooks/  
│   └── data_visualization.ipynb  
├── outputs/  
│   └── charts/  
│       ├── sales_by_category_bar.png  
│       ├── sales_by_year_line.png  
│       └── sales_distribution_pie.png  
├── src/  
│   └── visualization.py  
├── .gitignore  
├── LICENSE  
├── README.md  
└── requirements.txt
```

The notebook contains the complete interactive analysis, while `src/visualization.py` provides a reusable Python visualization script. The generated chart outputs are stored under `outputs/charts/`.

12. Conclusion

The Superstore Data Visualization project successfully demonstrates the use of Python, Pandas, and Matplotlib for business data analysis and visualization. The project converts raw sales data into clear visual representations and extracts useful business insights.

The work demonstrates practical understanding of data preprocessing, aggregation, visualization, and interpretation. The three required charts provide complementary perspectives: the bar chart compares category performance, the line chart shows the sales trend over time, and the pie chart communicates sales composition.

Overall, the project provides a concise example of how a structured business dataset can be transformed into understandable visual evidence that supports analytical reasoning and data-driven decision-making.

13. GitHub Repository

The project repository provided for the internship submission is:

https://github.com/ayeshafarooq566/DevSphere_Week2_Data_Visualization

The repository is intended to contain the project notebook, reusable visualization source code, generated chart outputs, and supporting project files described in this report.